

Autoclave Loading Using Deep RL

Victor U. Pugliese,* **Diogo S. Carvalho**,
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Outline

- **Motivation and Context**
- Reinforcement Learning
- Problem Formulation
- Environment & Solution
- Experimental results
- Insights & Conclusions

Motivation and Context

- Aerospace industry seeks lighter and stronger aircrafts;
- Composite materials are exceptional and widely adopted;

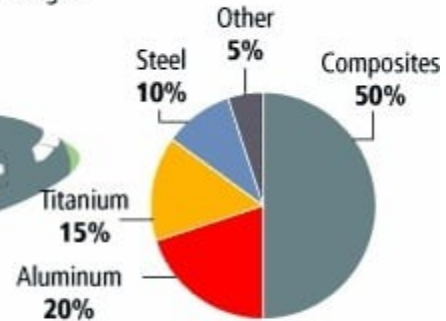
Materials used in 787 body

Fiberglass
Aluminum

Carbon laminate composite
Carbon sandwich composite
Aluminum/steel/titanium



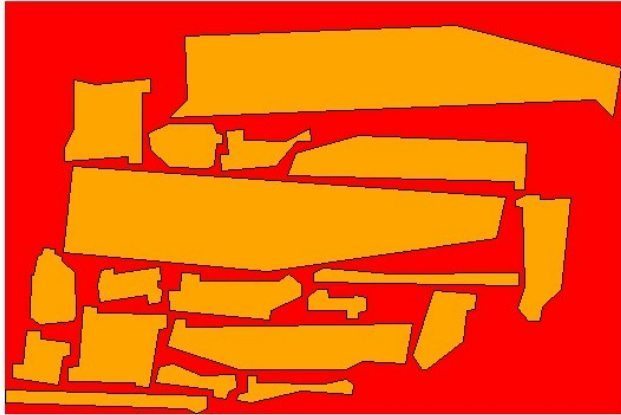
**Total materials used
By weight**



By comparison, the 777 uses 12 percent composites and 50 percent aluminum.

Motivation and Context

- Production involves cutting into preforms and curing in autoclaves;
- Solving a bin packing problem is critical for efficiency.



Motivation and Context

- 2D Rectangular bin packing (2DRBP) problem;

Paper	Algorithm	Actions
Zheng et al.	greedy + metaheuristic	item, coordinates, rotations, batch
Xie et al.	heuristic	item, rotation
Dios et al.	mixed integer linear programming	coordinates, rotation, cart, deck, tray

Motivation and Context

- Traditional solutions limited to **static** scenarios;
- Gap in dynamic environments, and **prioritization**.

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Reinforcement Learning

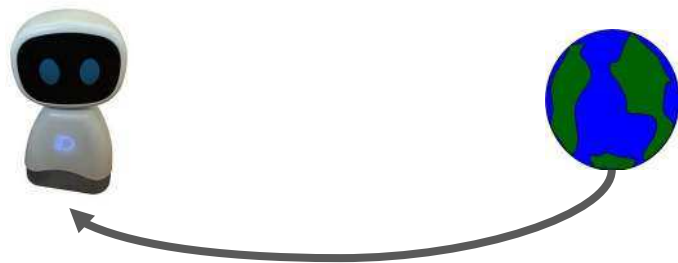
Reinforcement Learning



Reinforcement Learning

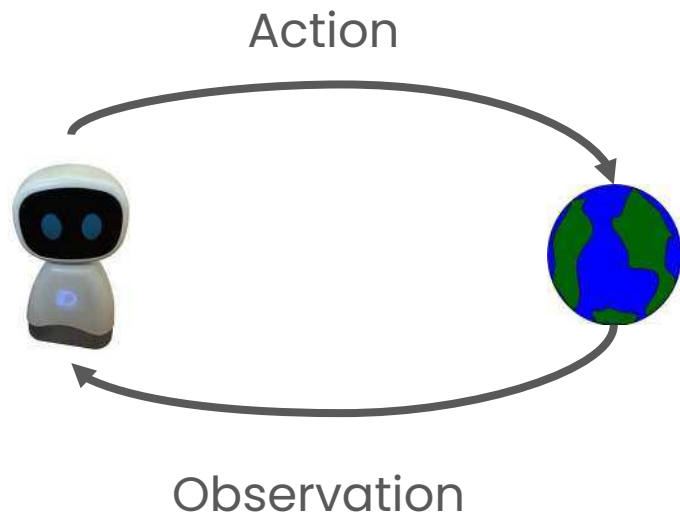


Reinforcement Learning

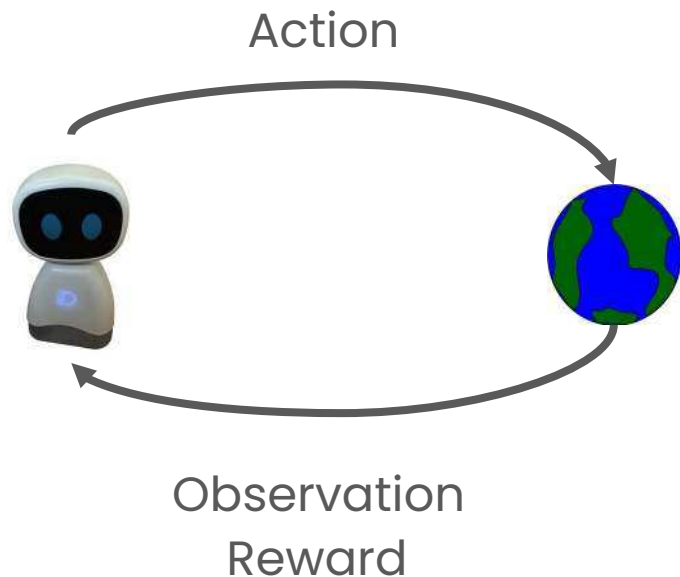


Observation

Reinforcement Learning



Reinforcement Learning



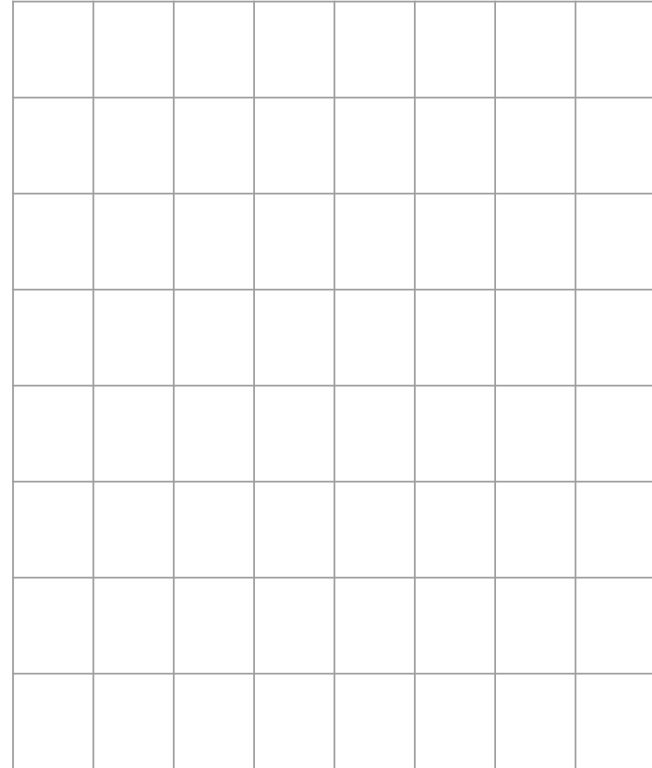
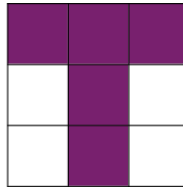
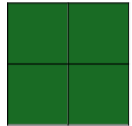
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Problem Formulation

Goal:

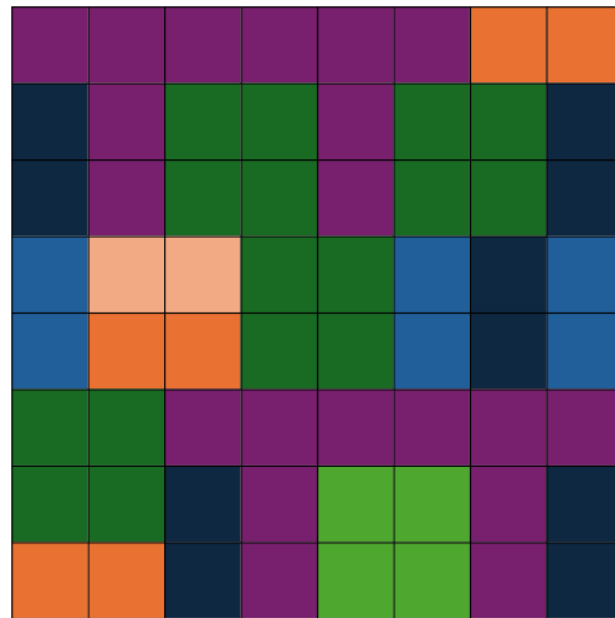
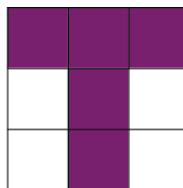
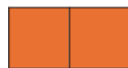
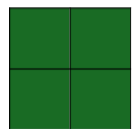
- Place items into the board;
- Maximize fill rate and prioritize.



Problem Formulation

Goal:

- Place items into the board;
- Maximize fill rate and prioritize.



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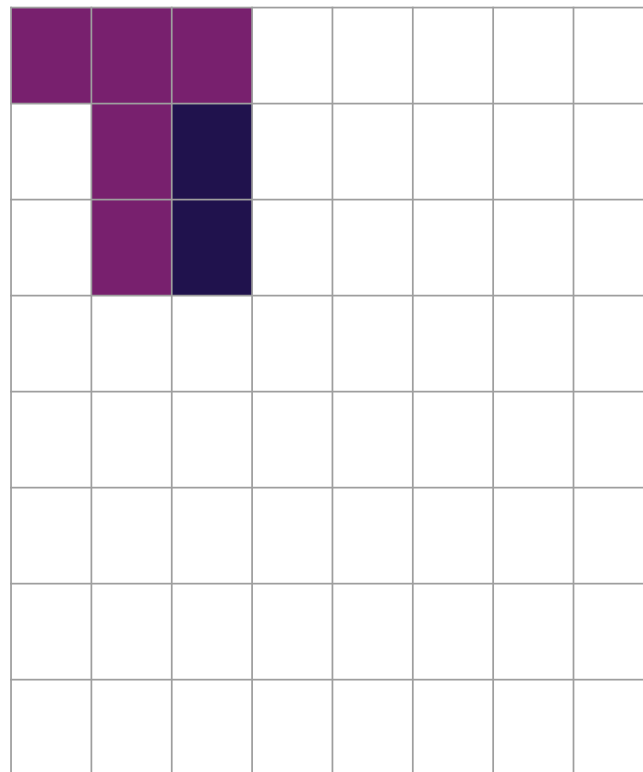
Environment

Environment

- Observation space:

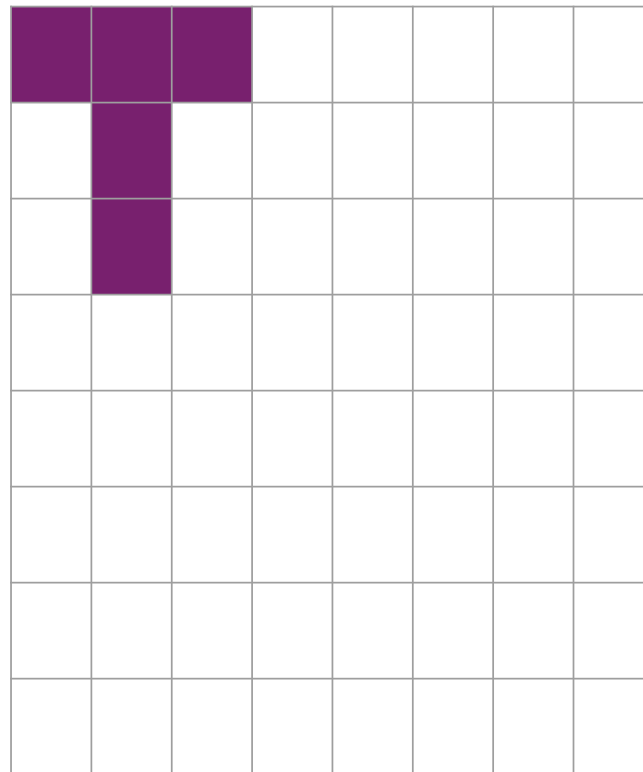
Environment

- Observation space:
 - Board at current time t ;



Environment

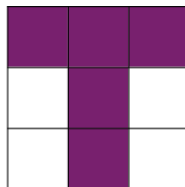
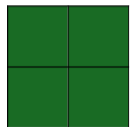
- Observation space:
 - Board at current time t ;
 - Board and piece at previous time $t-1$;



Environment

- Observation space:
 - Board at current time t ;
 - Board and piece at previous time $t-1$;
 - Counts of each piece type.

Piece



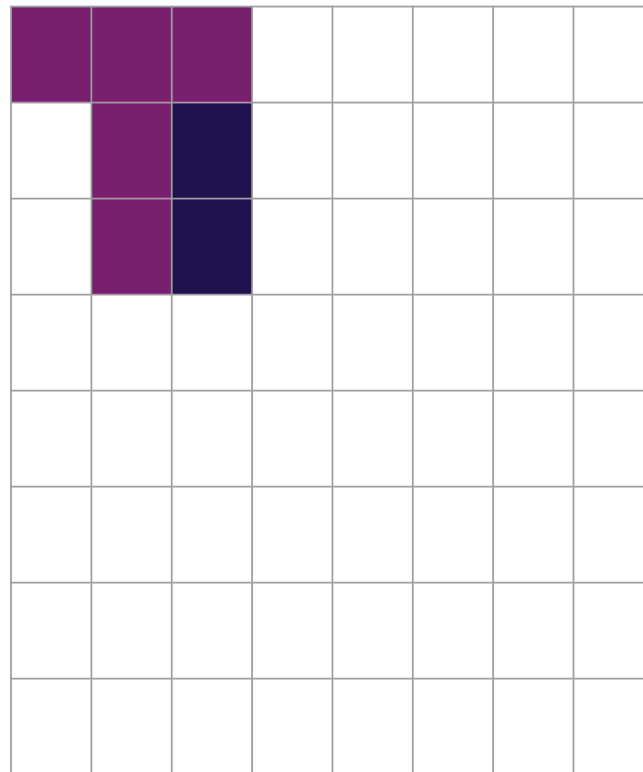
Count

5

4

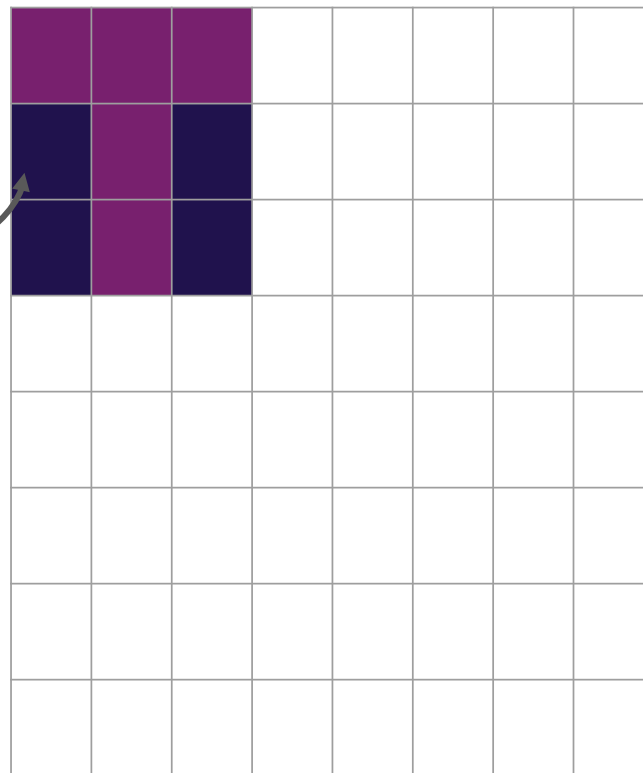
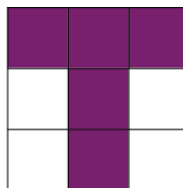
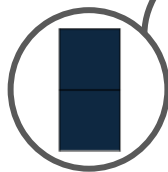
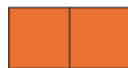
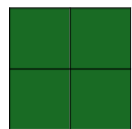
7

3



Environment

- Action space:
 - Item type;
 - Coordinates;

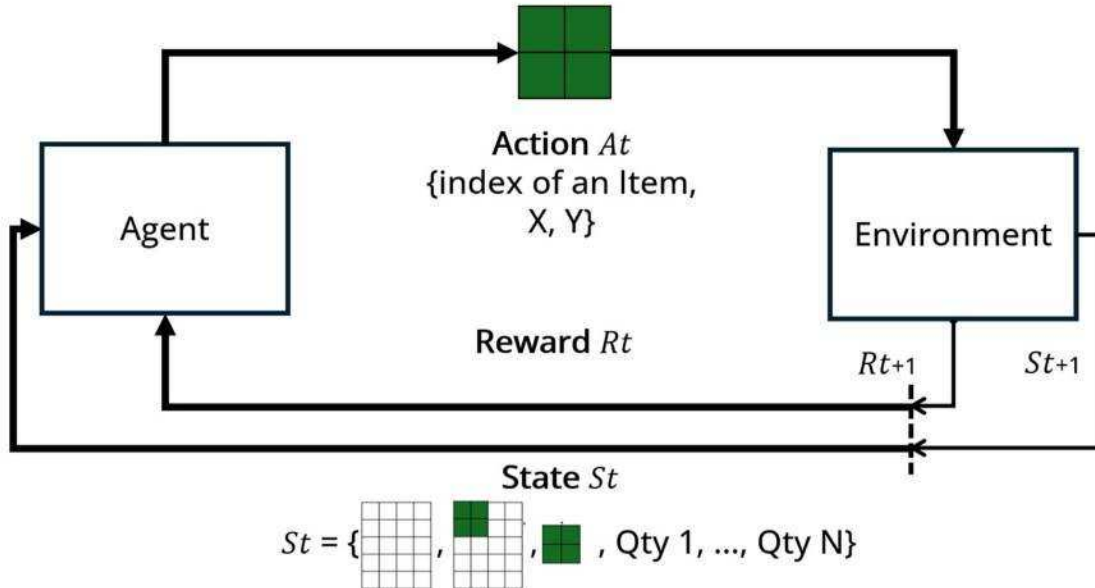


Environment

- Reward function:
 - + profit of placed item;
 - - penalty for invalid actions.

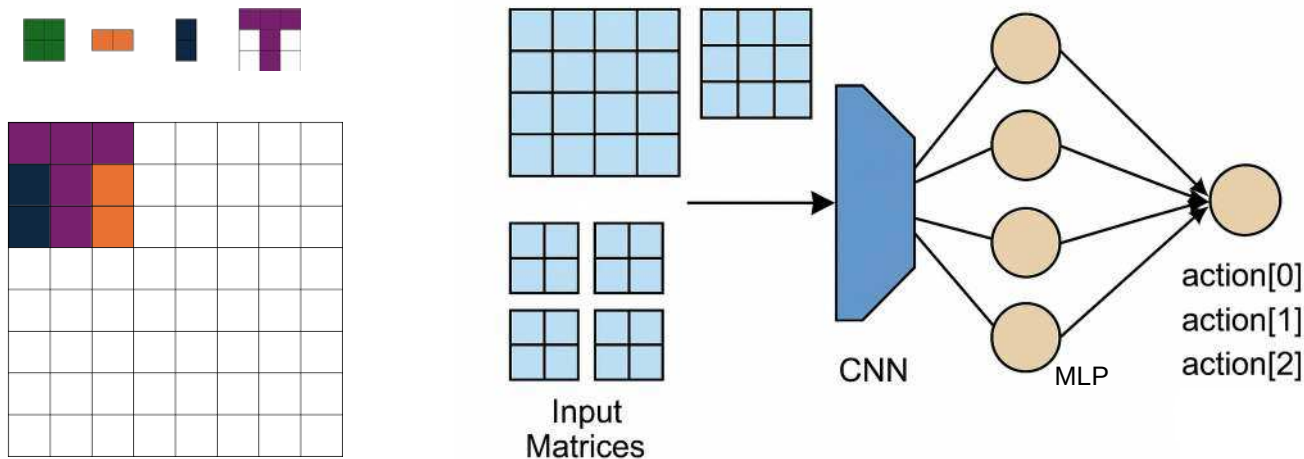
Environment

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Solution

- Proximal Policy Optimization (PPO) algorithm;
- Convolutional (CNN) and MLP layer processing.



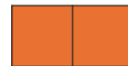
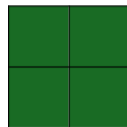
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Experimental Setup

Scenarios:

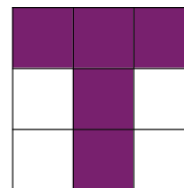
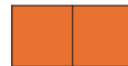
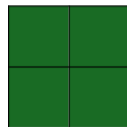
1. Only regular items;



Experimental Setup

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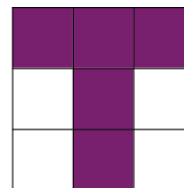
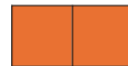
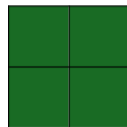
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2. Mixed regular and irregular items;



Experimental Setup

Scenarios:

1. Only regular items;
2. Mixed regular and irregular items;
3. Prioritizing high-value items:



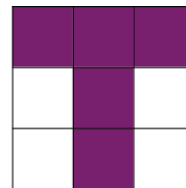
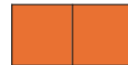
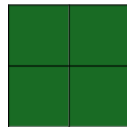
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Baseline:

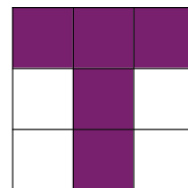
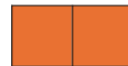
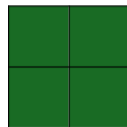
- Genetic algorithm.



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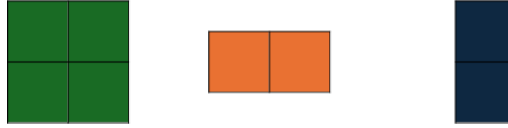
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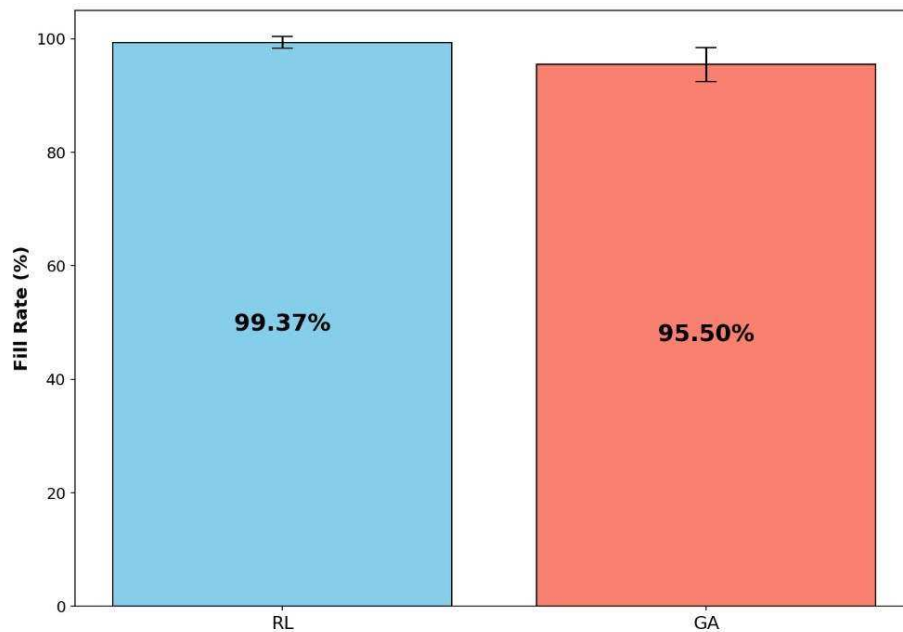
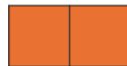
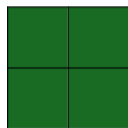
Metrics:

- Fill rate (%);
- Prioritized items placement rate (%).

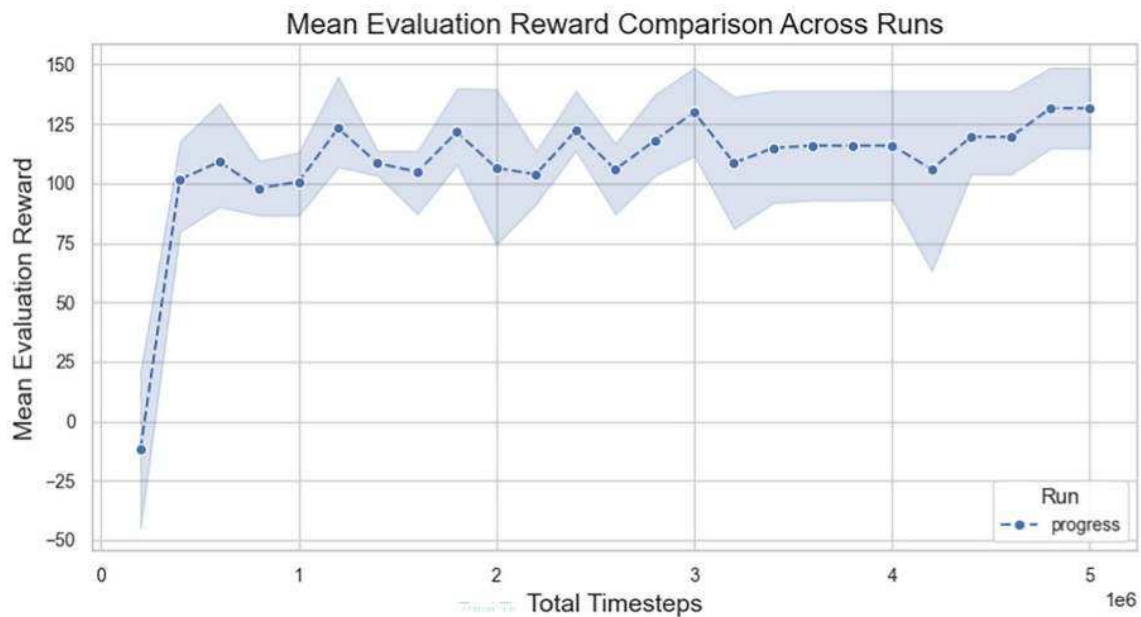
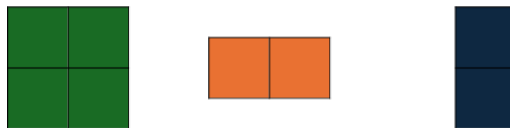
Case 1: Regular Items



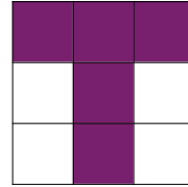
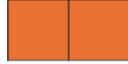
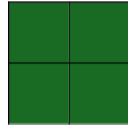
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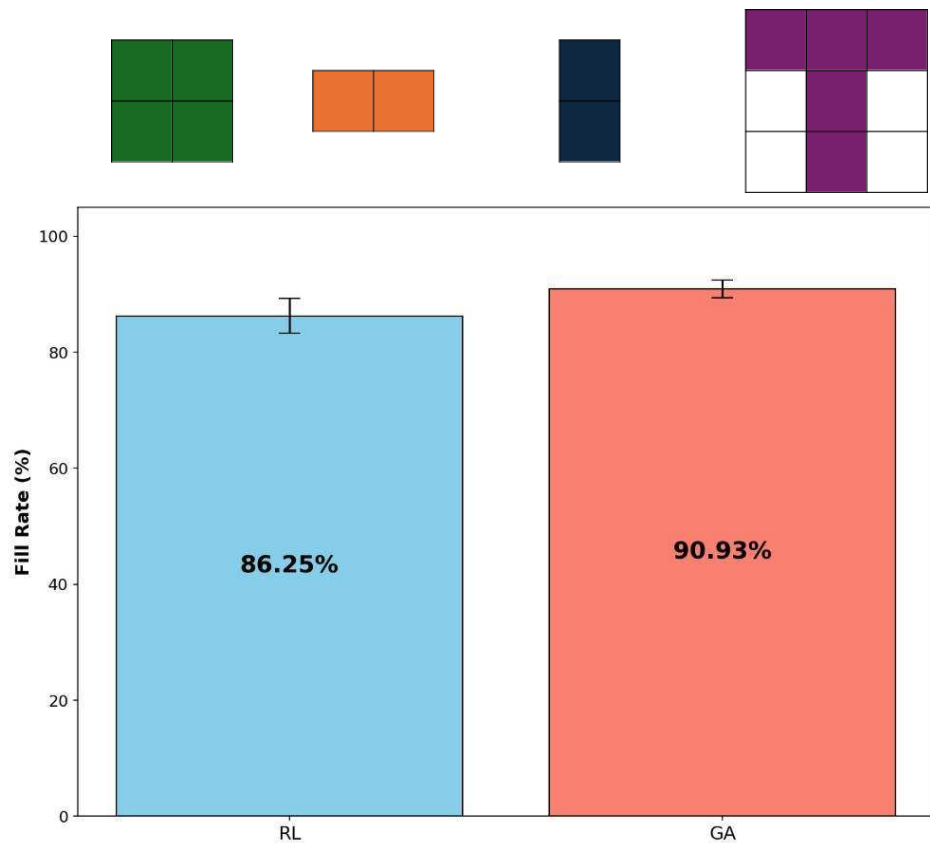
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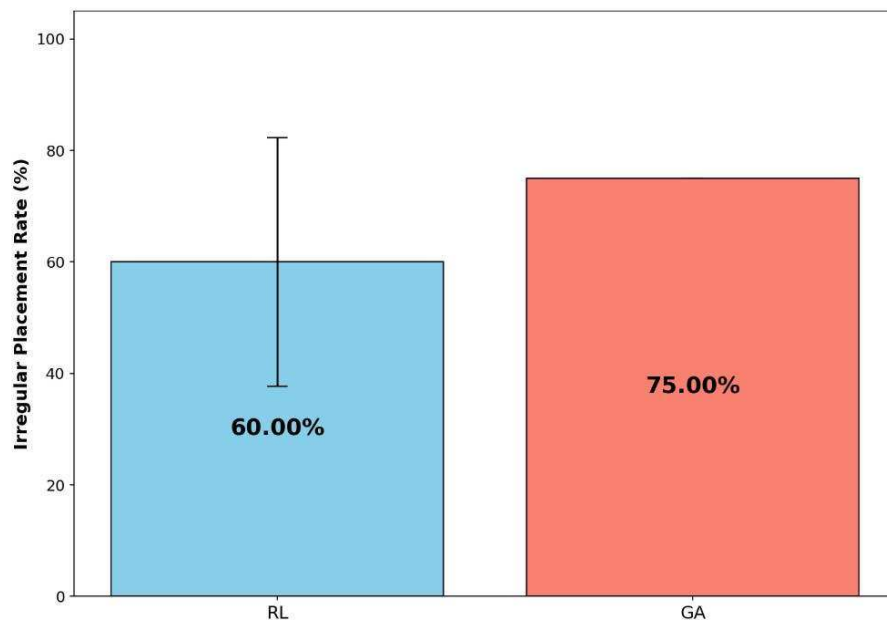
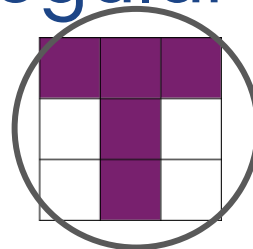
Case 2: Mixed Regular and Irregular Items



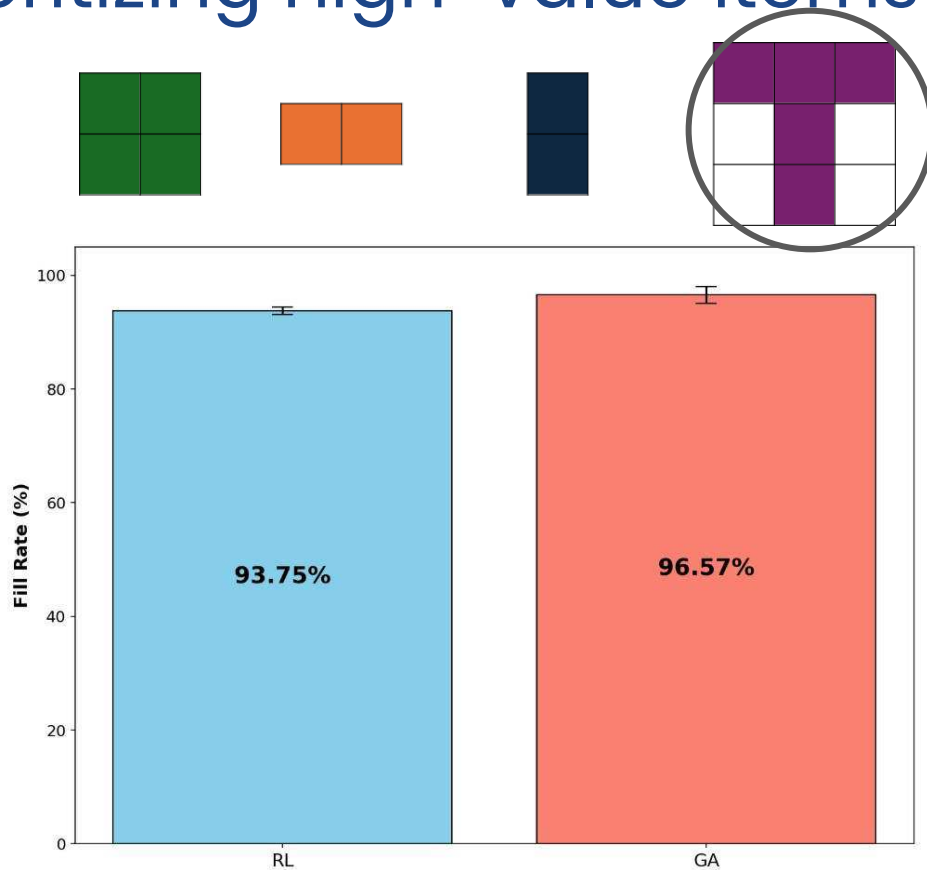
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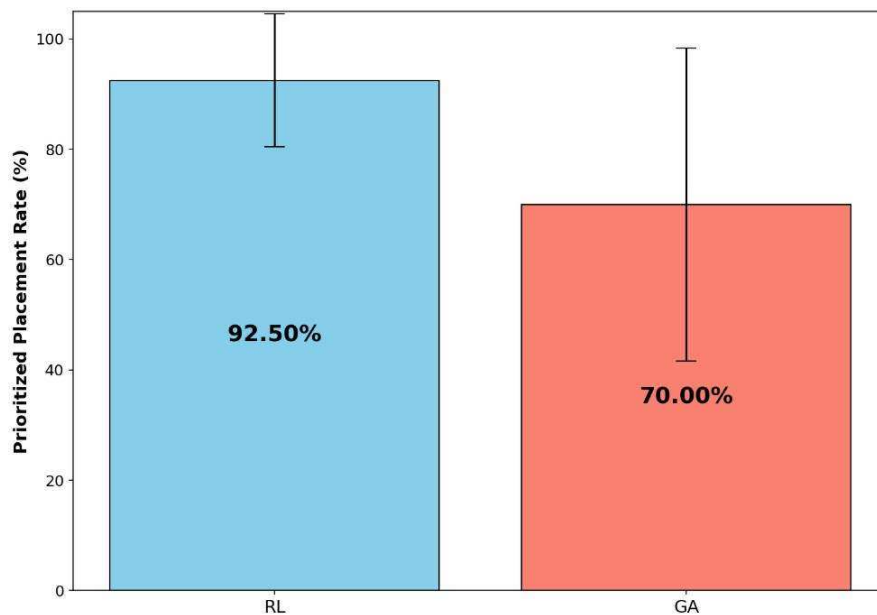
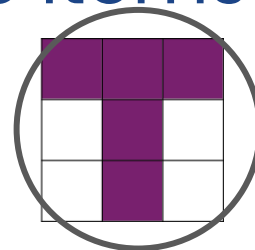
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Case 3: Prioritizing high-value items



Case 3: Prioritizing high-value items



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Analysis and Insights

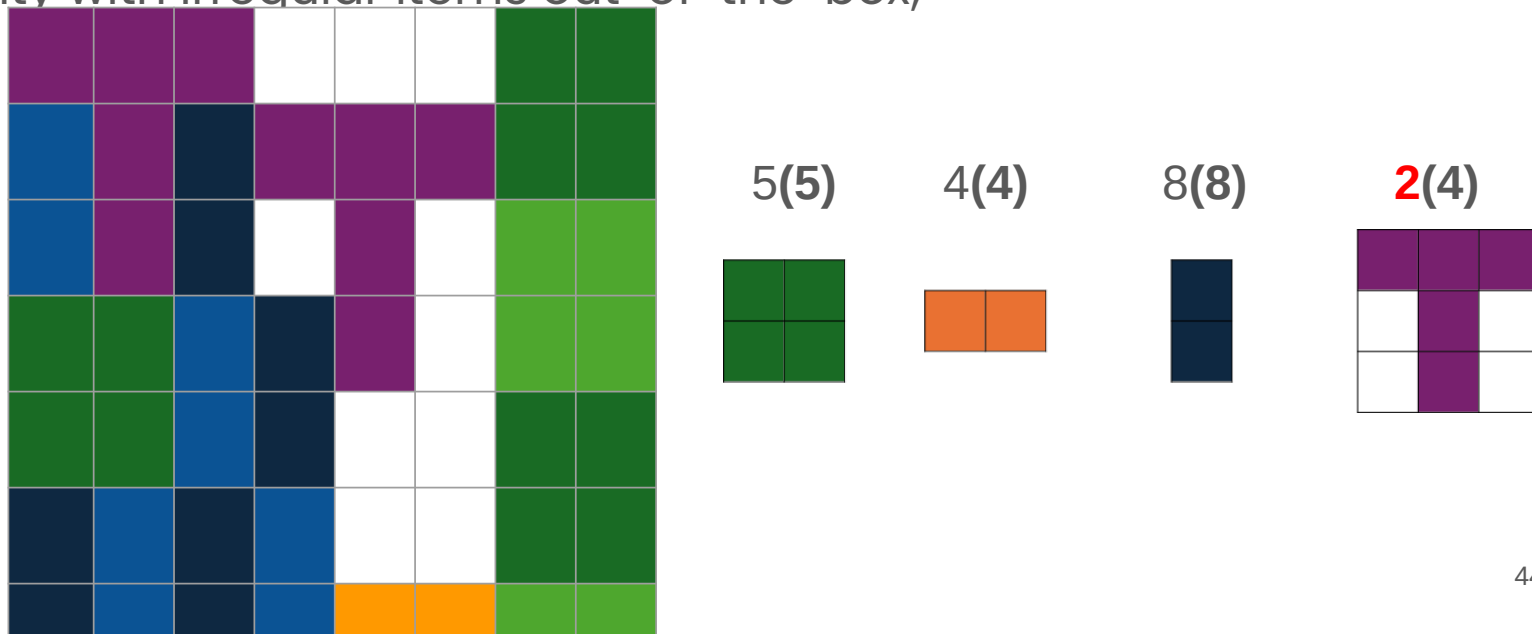
- RL is competitive with traditional methods;
- Allows flexibility in terms of **prioritization**.

Analysis and Insights

- RL is competitive with traditional methods;
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- Difficulty with irregular items out-of-the-box;

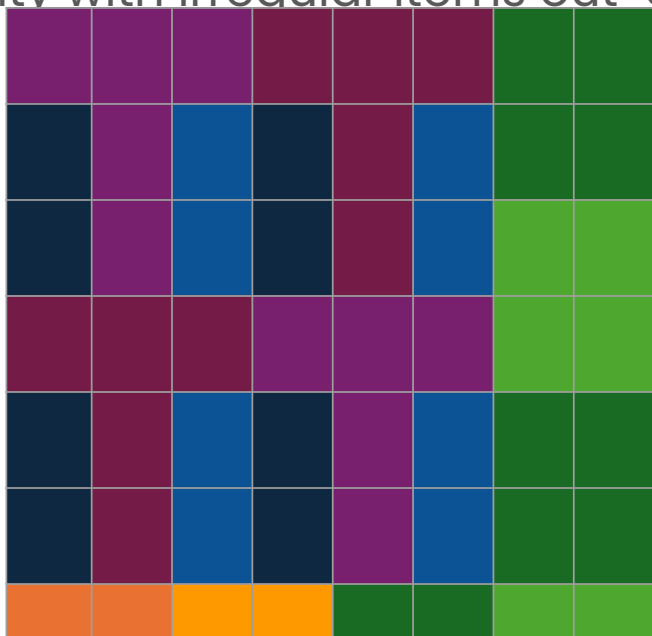
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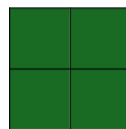


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5(5)



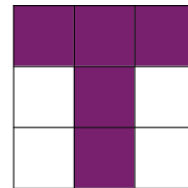
4(4)



8(8)



4(4)



Limitations and Future Work

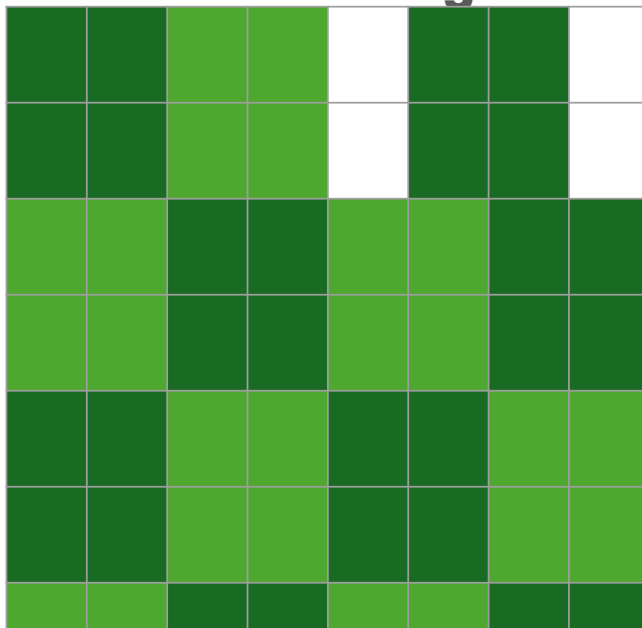
- Training time;

Limitations and Future Work

- Training time;
- **Key focus in generalizing item shapes.**
- Hybrid methods: RL + Heuristics to guide board completion

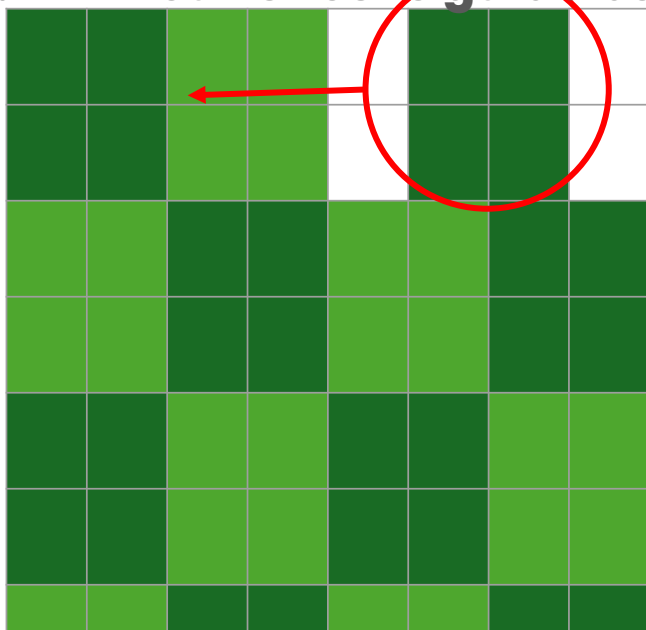
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- **Hybrid methods: RL + Heuristics to guide board completion**



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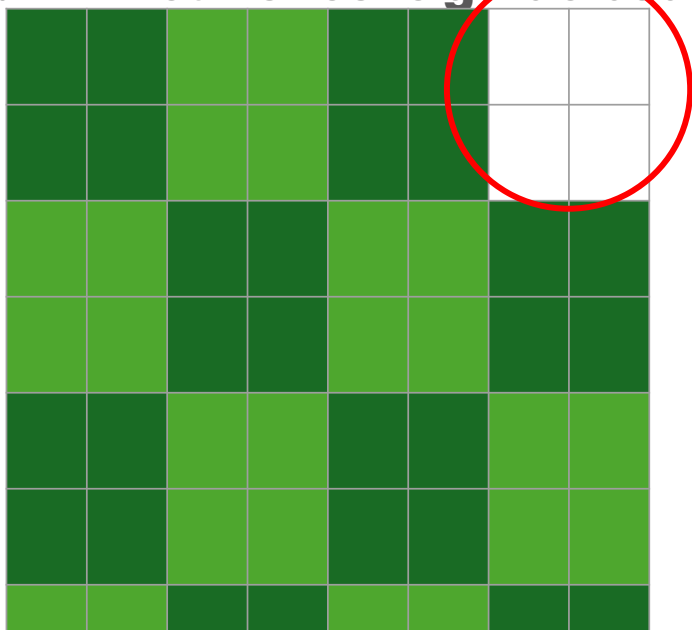
- Training time;
- Key challenge in generalizing item shapes.
- **Hybrid methods: RL + Heuristics to guide board completion**



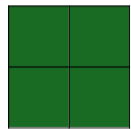
Bottom-left heuristic
allocating the item
as far to left as possible

Limitations and Future Work

- Training time;
- Key challenge in generalizing item shapes.
- **Hybrid methods: RL + Heuristics to guide board completion**



Now, we can add 1 item



Conclusions

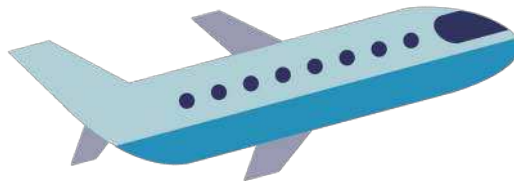
- Autoclave packing is a hard but relevant optimization task;
- RL is a feasible and promising solution;
- **+ enables priority-aware optimization beyond heuristics**

Conclusion

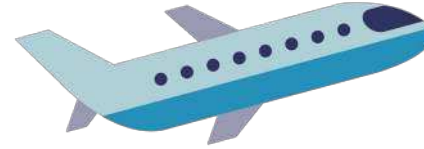
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Full paper



Thank you for your time!



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